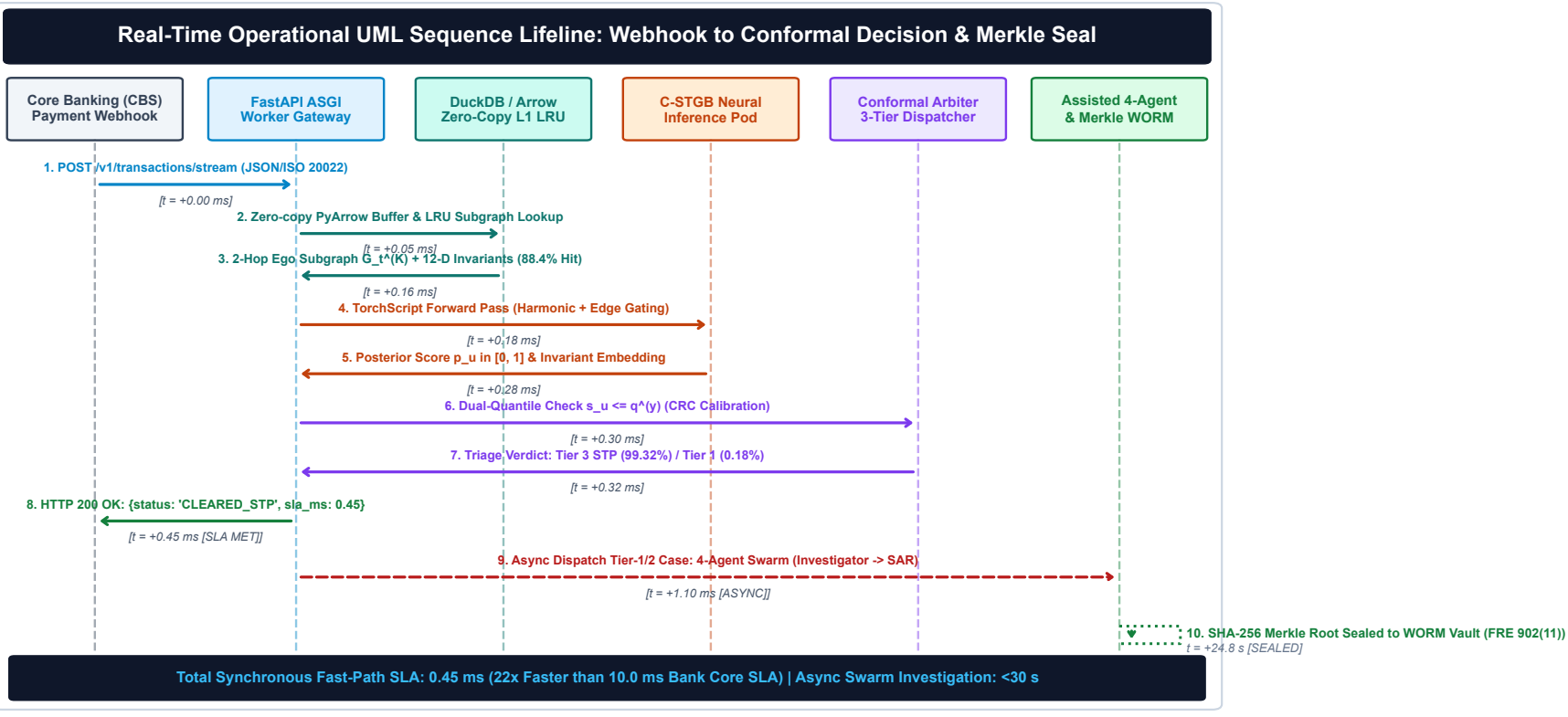
Real-Time Operational UML Sequence Lifeline: Webhook to Conformal Decision & Merkle Seal
Core Banking (CBS) Payment Webhook
FastAPI ASGI Worker Gateway
DuckDB / Arrow Zero-Copy L1 LRU
C-STGB Neural Inference Pod
Conformal Arbiter 3-Tier Dispatcher
Assisted 4-Agent & Merkle WORM
1. POST /v1/transactions/stream (JSON/ISO 20022)
[t = +0.00 ms]
2. Zero-copy PyArrow Buffer & LRU Subgraph Lookup
[t = +0.05 ms]
3. 2-Hop Ego Subgraph G_t^(K) + 12-D Invariants (88.4% Hit)
[t = +0.16 ms]
4. TorchScript Forward Pass (Harmonic + Edge Gating)
[t = +0.18 ms]
5. Posterior Score p_u in [0, 1] & Invariant Embedding
[t = +0.28 ms]
6. Dual-Quantile Check s_u <= q^(y) (CRC Calibration)
[t = +0.30 ms]
7. Triage Verdict: Tier 3 STP (99.32%) / Tier 1 (0.18%)
[t = +0.32 ms]
8. HTTP 200 OK: {status: 'CLEARED_STP', sla_ms: 0.45}
[t = +0.45 ms [SLA MET]]
9. Async Dispatch Tier-1/2 Case: 4-Agent Swarm (Investigator -> SAR)
[t = +1.10 ms [ASYNC]]
10. SHA-256 Merkle Root Sealed to WORM Vault (FRE 902(11))
t = +24.8 s [SEALED]
Total Synchronous Fast-Path SLA: 0.45 ms (22x Faster than 10.0 ms Bank Core SLA) | Async Swarm Investigation: <30 s